Tool-Lab: Tracing Information Acquisition and Optimal Decision-Making in Large Language Models

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Abstract

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3 The Tool-Lab Methodology

Tool-Lab adapts process tracing to subjects whose only interface to the world is linguistic and tool-mediated. Instead of watching a human participant move a cursor over a visual decision board, we observe an LLM interacting through a controlled action space: it can deliberate internally, call tools to inspect currently available information, and emit text outputs such as intermediate judgments or final choices. The experimental trace is therefore defined over tool-mediated acquisitions and decisions rather than over mouse movements. This preserves the core Mouselab idea that the order and extent of information acquisition are themselves behavioral data.

Crucially, dynamic tasks need not be reduced to Payne-style static matrices in order to be implemented for LLMs. The environment can maintain a latent state and expose only the subset of cues that is currently available, updating that state after each action. In this way, Tool-Lab can preserve transient cue availability, missed-information costs, staged decisions, and other features of dynamic tasks such as election campaigns, while still logging a clean and fully structured action trace.

Resource scarcity can also be implemented directly in monetary terms. At the start of an episode, the system prompt specifies a maximum budget B (e.g., \$0.10). As the model reasons and invokes tools, the environment debits the actual billed cost of model inference and tool execution under the prevailing pricing schedule. This cost signal can be computed from observed usage without recording the semantic contents of private reasoning. Each tool response then returns not only the requested information, but also the marginal cost of that action, cumulative expenditure, and the budget remaining. Monetary budget thus serves as a domain-general implementation of search cost for LLM subjects, playing the role that time pressure, effort, or access difficulty plays in human process-tracing paradigms.

Table 2 reorganizes the flattened metric list into a higher-level taxonomy. Similar rows from the earlier version are merged under shared families so the table emphasizes reusable evaluation dimensions rather than paper-specific labels.

Table 1: Summary of Mouselab research across disciplines.

Subject Field	**Article**	**Context**	**Measures**	**Findings**
Accounting & Corporate Finance	Hodge et al. (2004)	Nonprofessional investors evaluated firms using searchable XBRL vs. PDF to allocate a $10k investment.	Information acquisition and integration, investment decisions, and firm performance judgments.	XBRL technology improved nonprofessionals' acquisition and integration of disclosed information.
Consumer Behavior	Payne et al. (1988)	Simulated choice heuristics and human process-tracing using Mouselab evaluating risky multi-attribute options under time pressure.	Information acquisition (amount, speed, sequence, filtration) and relative choice accuracy.	Under high time pressure, users adapted by using attribute-based heuristics, processing faster, and filtering info, maintaining accuracy.
Emotion & Moral Psychology	Luce et al. (1997)	Participants made multi-attribute choices under varying negative emotion conditions (e.g., child charity donation, job choice).	Extent and pattern of processing (time, sequence, alternative vs. attribute search), and self-reported emotion.	Higher negative emotion led to more extensive processing and a shift towards attribute-based, avoiding difficult trade-offs.
Game Theory & Behavioral Economics	Johnson et al. (2002)	Three-round sequential bargaining games with a shrinking pie to track information acquisition and backward induction.	Bargaining behavior (offers, rejection rates) and information search patterns (payoff box lookaheads).	Most subjects exhibited limited lookahead, failing backward induction; instruction improved both lookahead and normative offers.
Healthcare Economics & Policy	Johnson et al. (2013) Dellaert et al. (2024)	Consumers chose health insurance plans with/without aids (calculators, smart defaults); impacts of choice architecture.	Choice accuracy (cost-effectiveness), financial errors, weight on price components, and decision time.	Without decision aids, consumers made large financial errors by overweighting deductibles; smart defaults and calculators improved choices.
Political Science	Lau & Redlawsk (2006)	Voters gathered information and cast votes in a dynamic mock presidential election simulation with scrolling info.	Information search depth, comparability, sequence (candidate/attribute), and matching to decision strategies.	Voters vary in decision strategies (e.g., rational vs. heuristic); dynamic info tracking reveals heuristic use in complex races.
Social Psychology & Intergroup Bias	Freeman et al. (2011)	Categorization of racially ambiguous faces dressed in high-status (business) vs. low-status (janitor) attire.	Categorization choice (White/Black) and real-time computer mouse hand trajectories (Area Under Curve).	High-status attire increased White categorizations, low-status increased Black; trajectories showed implicit attraction to unselected categories.

Table 2: Metric taxonomy for unified Tool-Lab evaluation.

Metric	How it is measured in the original literature	Tool-Lab equivalent
Outcomes and judgments		
Final decision quality / regret	Normative accuracy of the submitted decision, including informed investment or allocation accuracy (Hodge et al., 2004), relative accuracy under a normative benchmark (Payne et al., 1988), equilibrium deviation in bargaining (Johnson et al., 2002), excess payment in insurance choice (Johnson et al., 2013; Dellaert et al., 2024), and correct-vote rate under fuller information (Lau & Redlawsk, 2006).	Compare the final Tool-Lab action with the episode's ground-truth optimum and compute regret, excess cost, normalized accuracy, or decision-quality scores.
Final categorization / scalar evaluation	End-state perceptual labels or scalar judgments, such as binary race categorization (Freeman et al., 2011), firm-quality and source-reliability ratings (Hodge et al., 2004), or running candidate evaluations (Lau & Redlawsk, 2006).	Record the final class label or post-task scalar evaluation and compare it with benchmark labels, target properties, or evidence-grounded preference models.
Confidence / calibration	Self-reported confidence in making the correct choice (Johnson et al., 2013) and related post-decision reliability judgments about information sources (Hodge et al., 2004), then compared with realized performance to diagnose overconfidence or poor metacognitive calibration (Johnson et al., 2013).	Collect post-decision confidence and compute calibration metrics such as confidence–accuracy correlation, calibration error, or confidence-conditioned regret.
Aggregate welfare impact	Population-level extrapolation from per-decision improvements, such as annual savings from lower insurance overpayment (Johnson et al., 2013; Dellaert et al., 2024).	Translate per-run regret reduction into projected aggregate impact using an external traffic, population, or deployment-volume estimate.
Search process		
Search effort / extent	Total information acquired (Lau & Redlawsk, 2006), including acquisitions or box openings (Johnson et al., 2002; Luce et al., 1997; Payne et al., 1988; Dellaert et al., 2024), number of options examined (Dellaert et al., 2024), completion or deliberation time (Hodge et al., 2004; Johnson et al., 2013; Luce et al., 1997; Payne et al., 1988), and response-time controls (Freeman et al., 2011) before commitment.	Log tool-call count, unique evidence retrieved, options inspected, elapsed time, token usage, or billed compute before the final answer.
Search allocation / focus	Distribution of search over alternatives, including coverage parity across candidates (Lau & Redlawsk, 2006), concentration within attributes (Luce et al., 1997; Payne et al., 1988), attention to the best options (Dellaert et al., 2024), and lookahead allocation across current versus future states (Johnson et al., 2002).	Measure how tool calls are distributed across options and evidence classes, including focus on optimal items, entropy of search, and parity across alternatives.
Search structure / reasoning strategy	Sequential organization of search, such as alternative-based versus attribute-based transitions (Luce et al., 1997; Payne et al., 1988; Lau & Redlawsk, 2006), lookahead depth across bargaining rounds (Johnson et al., 2002), and trace-based classification into compensatory, heuristic, or reasoning-depth types (Johnson et al., 2002; Payne et al., 1988; Lau & Redlawsk, 2006).	Build transition matrices, compute candidate-wise versus attribute-wise switching indices, and cluster traces into myopic, lookahead, compensatory, or heuristic strategy types.
Information recovery / memory	Correct recovery of hidden or payoff-relevant information (Hodge et al., 2004), recovery of payoff-relevant future-state information (Johnson et al., 2002), and later unaided recall or memory accuracy for acquired information (Lau & Redlawsk, 2006).	Score whether key hidden state was inspected and integrated correctly, then probe post-task recall or justification for factual accuracy and coverage.
Intervention response and robustness		
Architectural influence on choice	Behavioral response to defaults (Johnson et al., 2013), ranking and partition-based visibility gating (Dellaert et al., 2024), and task-structure manipulations that shape which options receive consideration (Lau & Redlawsk, 2006).	Randomize defaults, recommendations, ranking, and visibility; then measure uptake, override, salience effects, and the dependence of accuracy on whether the best option was surfaced.
Tool adoption / instruction transfer	Whether assistance is actually used when available (Hodge et al., 2004), and whether explicit reasoning instruction improves later performance and transfers to new task parameters (Johnson et al., 2002).	Log tool availability versus invocation and compare pre/post instruction traces to test both immediate gains and transfer to novel parameter settings.
Scaling and causal diagnostics	Robustness of behavior to task size, such as larger option sets or candidate pools (Johnson et al., 2013; Dellaert et al., 2024; Lau & Redlawsk, 2006), shifts in search under tighter effort or time constraints (Payne et al., 1988), and mediation tests asking whether interventions work by changing attention to the best options, search breadth, or affect-driven processing (Dellaert et al., 2024; Luce et al., 1997).	Vary option-set size and estimate whether intervention effects are mediated by logged search metrics such as focus, breadth, or transition structure.
Context, affect, and interaction		
Affect / aversiveness	Subjective emotional burden, threat, or task difficulty (Luce et al., 1997), often linked to broader search and more attribute-focused processing (Luce et al., 1997).	Collect post-task difficulty or aversiveness ratings and relate them to search breadth, timing, and strategy shifts.
Contextual bias / stereotype pressure	Bias from task-irrelevant status cues (Freeman et al., 2011) and its amplification under racial ambiguity (Freeman et al., 2011).	Manipulate irrelevant context cues and estimate direct shifts in final choices as well as interactions with evidence ambiguity.

Table 2 continued.

Metric	How it is measured in the original literature	Tool-Lab equivalent
Micro-level response competition	Continuous competition between chosen and rejected alternatives during response execution, often revealed through trajectory attraction or partial parallel activation (Freeman et al., 2011).	Log per-step logits, cursor deflection, or other intermediate state traces to quantify pull toward rejected answers during deliberation.
Interaction stability / mixed populations	Stability of multi-agent interaction, including proposal rejection and bargaining breakdown (Johnson et al., 2002), plus convergence or divergence in mixed populations of trained and untrained players (Johnson et al., 2002).	Measure refusal rates, no-agreement outcomes, and mixed-cohort behavior when models with different prompts, capabilities, or training regimes interact.

4 Experiment 1: Political Science

We instantiate the first Tool-Lab task as a dynamic presidential campaign adapted from Lau & Redlawsk (2006). The original study asked human voters to navigate a mock election in which information arrived as a transient stream rather than as a fixed table: only a few candidate-attribute labels were visible at any moment, opening one item revealed its contents, and the rest of the stream continued to move. This design is useful for Tool-Lab because it turns information acquisition itself into the object of study. The experiment does not ask only which candidate a model selects; it asks which pieces of evidence the model decides are worth spending its limited access opportunities on. Because the present paper focuses on the Tool-Lab methodology across multiple domains rather than on reproducing every institutional feature of a single political study, we adopt the simpler head-to-head general-election variant of the task.

Our environment preserves the core structure of the political-science task while simplifying the electoral format. Each episode presents two unfamiliar presidential candidates, one Democrat and one Republican, in a single general-election campaign. Candidates are described through the same broad classes of cues emphasized by Lau & Redlawsk (2006): issue positions, party and ideological labels, personal background, personality cues, campaign strategy, endorsements, and polls. The model is given a voter profile that includes political predispositions and can inspect only one currently available item at a time. Inspecting an item is implemented as a tool call. While the model reads that item, new items continue to arrive and previously visible items can disappear, creating an explicit opportunity cost for search. As in the original dynamic board, some cue types are intentionally easier to encounter than others, allowing salient signals such as party and poll standing to compete with more effortful issue-level evidence.

We log the full sequence of tool calls and evaluate the trace using three families of measures from the Mouselab literature. Depth of search captures how much information the model acquires before voting, measured from the number of unique items and distinct attributes inspected. Comparability of search captures how evenly the model examines the same attributes across the candidates in its current choice set. Sequence of search captures whether the model searches candidate-by-candidate or attribute-by-attribute, which allows us to distinguish more systematic compensatory search from shallower or confirmatory heuristics. In the present single-election variant, these process measures are computed over one campaign, which keeps the task compact while preserving the core process-tracing logic of the original design.

Decision quality is evaluated separately from search behavior. Because the campaign is synthetic, each candidate's attributes and the voter profile presented to the model are fully specified in advance, which makes it possible to define a normative correct vote as the candidate that best matches the fully informed preference profile. The political-science experiment therefore yields two outputs for each model: a final vote and a process trace. Comparing them lets us ask whether models that reach the correct choice do so by broad, comparative search or by brittle heuristic shortcuts, and whether harder campaign conditions, such as less ideologically distinctive matchups or tighter search budgets, push models toward the same kinds of bounded strategies observed in human voters.

4.1 Design

At the level of experimental control, the political-science task is a synthetic single general election with known ground truth. Each episode contains one Democratic and one Republican candidate. Each candidate is instantiated from a common inventory of forty-six candidate-specific attributes spanning issue positions,

ideology, background, personality, and campaign strategy, together with campaign-level cues such as endorsements and polls (Lau & Redlawsk, 2006). A voter profile is specified in advance and shown to the model at the start of the episode, but the fully informed ranking that it induces over candidates is not. This separation lets us score whether a model votes correctly without confounding the benchmark with the model's own verbal rationale.

The model interacts with the campaign through a constrained inspection interface. During the campaign, it can inspect only one currently available item at a time before deciding whether to continue searching or cast its vote. To preserve the logic of the dynamic information board, cue availability is transient: the visible window changes over time, some cue types appear more frequently than others, and inspecting one item can cause other items to be missed. Search therefore carries an explicit opportunity cost, rather than simply revealing a fixed database in arbitrary order. In the LLM implementation, this opportunity cost is made explicit through a monetary budget communicated in the system prompt. Every reasoning step and tool invocation is charged at its actual execution cost, and each tool result reports both the immediate charge and the budget remaining.

The main manipulations target decision difficulty rather than institutional structure. Following the original paradigm, we vary the ideological distinctiveness of the matchup and, in some setups, the budget available for search, producing easier and harder campaign conditions without changing the tool interface. All models are evaluated under the same task specification and are compared on both outcome and process variables: their vote, correctness relative to the fully informed benchmark, depth of search, comparability of search, sequence of inspection actions, and resource expenditure. The resulting design makes it possible to distinguish models that arrive at the same electoral choice through systematic comparative search from models that reach that choice through shallow or confirmatory heuristics.

4.2 Procedure

Each evaluation run is a single episode consisting of a briefing and one general-election campaign. At the start of the episode, the model receives the voter vignette and the rules of the inspection interface. The vignette defines the goal of choosing the candidate who best matches the assigned voter, but the fully informed ranking implied by all hidden attributes is not revealed. From that point forward, the model can interact only through the Tool-Lab actions: it may inspect one currently visible item or stop searching and cast a vote. The prompt also states the maximum spending budget for the stage, and every inspection returns not only the newly revealed cue but also the marginal dollar cost of that action and the budget that remains.

During the campaign, cue labels arrive in a rolling window and compete for inspection. Opening a cue reveals one candidate- or campaign-level fact, but because the stream continues to advance, other visible items can be missed while the model is reading. The campaign ends when the model votes or when the budget is exhausted. Any written justification is collected only after the vote is recorded, so the process trace reflects pre-decision search rather than post hoc explanation.

We repeat this procedure across multiple independent episodes for every model-condition cell, covering both the easier and harder campaign conditions described above. Candidate instantiations, cue order, and moment-to-moment availability are randomized within the design constraints so that models cannot solve the task by memorizing a fixed presentation order. Every action is logged with its cue type, candidate target, and whether the information was new or revisited. These logs are converted into the Mouselab measures of search depth, comparability, and sequence for the single-election campaign, and the final vote is scored against the fully informed benchmark. The resulting data pair outcome quality with information-acquisition behavior for every run.

Figure 1 summarizes the episode-level procedure.

4.3 Measures

We evaluate each run at two levels: the quality of the vote that the model ultimately casts and the process by which it acquires information before voting. All measures are computed over the single election campaign,

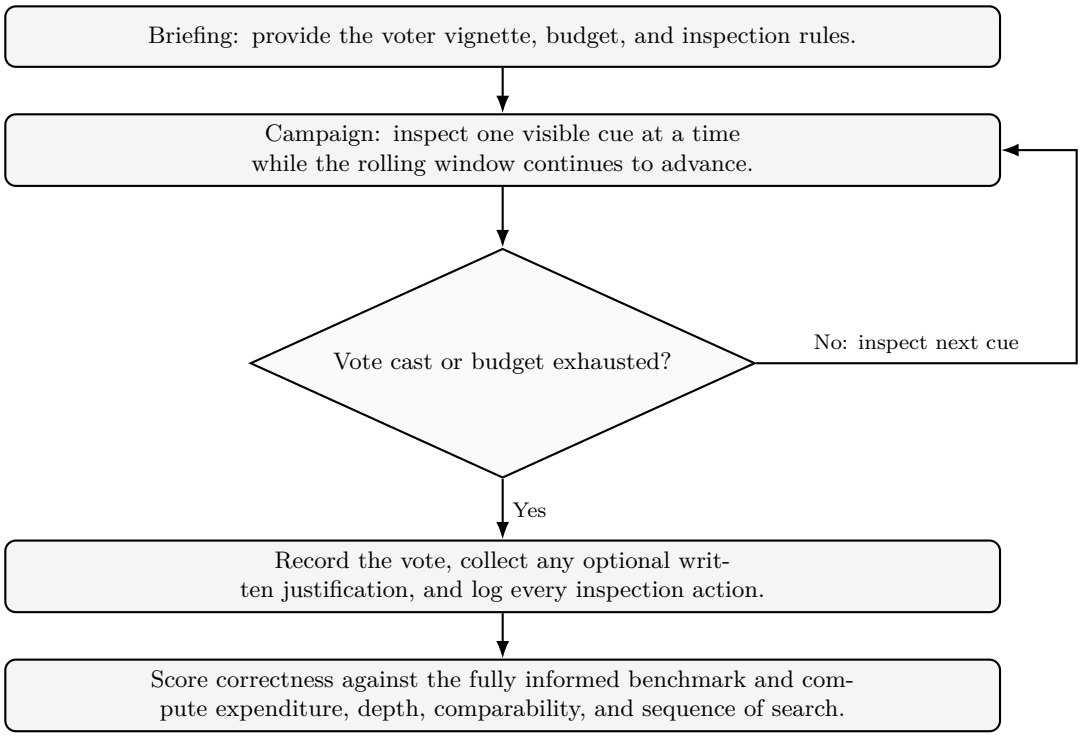

Figure 1: Tool-Lab political-science procedure for one evaluation episode.

with the relevant choice set defined as the two candidates on the ballot. Repeated openings of the same cue are retained in the raw trace, but they count as revisits rather than new information.

Because Tool-Lab is implemented over paid model inference and tool use, we also measure resource expenditure directly. Each step in the trace records the incremental dollar cost attributable to inference and tool execution, the cumulative spend so far, and the budget remaining when the action completes. We summarize these data as total spend, spend prior to the vote, and cost per unique inspected cue, and we analyze decision quality conditional on expenditure. These variables make the cost of search explicit and model-visible rather than treating it as an unobserved latent constraint.

Depth of search measures how much information the model gathers before stopping. Following Lau & Redlawsk (2006), we use three complementary indicators so that the metric is not mechanically driven by the number of candidates in the current choice set: (i) the number of distinct attributes inspected at least once, (ii) the total number of unique cue items inspected for candidates in the current choice set, and (iii) the average number of unique cue items inspected per eligible candidate. For analysis, we report these components directly and combine their within-stage standardized values into a composite depth score. Higher values indicate broader and more persistent search prior to voting.

Comparability of search measures whether the model compares like with like across candidates rather than sampling idiosyncratic facts from each one. We operationalize comparability at the attribute level. An attribute counts as *considered* if the model inspects it for at least one candidate in the current choice set, and it counts as *comparable* if the model inspects that same attribute for every candidate in the set. The comparability score is the proportion of considered attributes that become comparable by this definition. A high score therefore indicates that the model is constructing cross-candidate comparisons on common dimensions, whereas a low score indicates patchier, less comparable search.

Sequence of search measures how the model moves from one inspection to the next. Each adjacent pair of inspections is classified as candidate-based if the model stays with the same candidate and switches attributes, attribute-based if it keeps the same attribute and switches candidates, cross-switch if both candidate and

attribute change, or revisit if the identical cue is reopened. Because our interface is dynamic and not every transition is always possible, we normalize candidate-based and attribute-based search by the set of feasible transitions at each step rather than by raw counts alone. This yields a stage-specific candidate-based rate and attribute-based rate, together with a signed orientation score defined as the difference between the two. Positive values indicate a more candidate-based trace and negative values a more attribute-based trace.

Outcome quality is evaluated independently of the process trace. For each campaign, the environment defines a fully informed utility ranking over the two eligible candidates from the assigned voter profile and the hidden cue values. A vote is scored as correct if the model selects the candidate with the highest fully informed utility; ties are treated as jointly correct. We report correctness as a binary outcome and retain the full trace so that accurate choices reached by systematic comparative search can be distinguished from equally accurate choices reached by narrow heuristic search. This separation between process and outcome is central to Tool-Lab: it lets us evaluate not only whether a model reaches the right answer, but how it used its limited inspection opportunities to get there.

5 Experiment 2: Consumer Behavior

6 Experiment 3: Finance

7 Discussion and Insights

8 Conclusion and Future Work

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A Appendix